INCH-POUND
MIL-PRF-3098/111E
11 July 2011
SUPERSEDING
MIL-PRF-3098/111D
27 July 2005

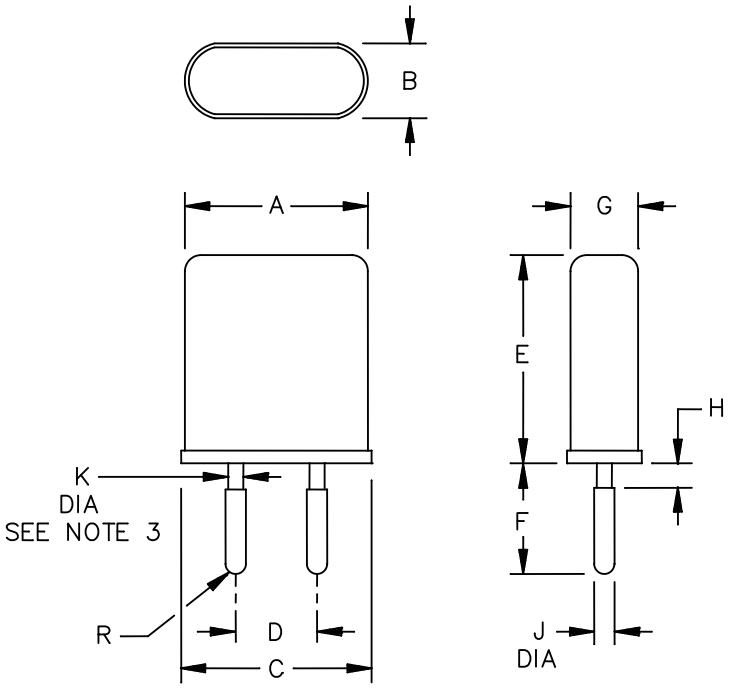
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR8/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 1 MHz to 10 MHz; fundamental; noncontrolled; series resonance.



Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.725	---	18.42
B	---	.352	---	8.94
C	---	.757	---	19.23
D	.460	.476	11.68	12.09
E	---	.775	---	19.69
F	---	.448	---	11.38
G	---	.317	---	8.05
H	.012	.047	0.30	1.19
J	.091	.095	2.31	2.41
K	.048	.052	1.22	1.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR8/U.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:53Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 1 MHz to 10 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Operating temperature ranges (noncontrolled): -40°C to 70°C, inclusive.

Frequency tolerances (operating temperature range): ±50 ppm.

Equivalent resistance: See table I.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
1.0+ to 2.0	450
2.0+ to 3.0	250
3.0+ to 4.0	150
4.0+ to 5.0	100
5.0+ to 6.0	75
6.0+ to 7.0	55
7.0+ to 10.0	50

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Part or Identifying Number (PIN): For Product Level B, PIN shall be CR8/U or CR8B/U (followed by specified frequency). For Product Level S, PIN shall be CR8S/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-PRF-202

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:53Z
Check the source to verify that this is the current version before use.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MD, MI, SM
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2011-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:53Z
Check the source to verify that this is the current version before use.